ST JOSEPH ENGINEERING COLLEGE

An Autonomous Institution

Department of Electronics and Communication Engineering

DRY SOLAR PANEL CLEANING SYSTEM

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Introduction

- Solar energy is a great way to generate power, and it's expected to make up 10% of our energy mix by 2030.
- Unfortunately, dust on solar panels can reduce their effectiveness by 30%. We also waste a lot of water cleaning them, with around 10-30 billion gallons used worldwide every year.
- More than 10 billion gallons of water is used to clean panel annually.
- To solve this problem, we've developed a new way to clean solar panels without using any water.

Introduction





Rooftop solar panels

Introduction





Dust Accumulation on Solar Panels Impacting Efficiency

Literature Review

Sl. No.	Paper title	Author name	Year	Key Points	Gaps Identified
1	Automatic solar panel cleaning using wiper.	Innico.R,Karthish Durai.A.M, Poovalingam	2023	Combination of wiper and DC motors for cleaning.	Wiper creates scratches on panel.
2	Cleaning of Panels Utilizing the Downward Thrust of a Drone.	Electrical Engineering Department	2020	Dust removal using airflow generated by its propellers.	Require skilled profession to handle.
3	Automatic solar panel cleaning system based on Arduino for dust removal.	Md. Rawshan habib, Md Tanvir	2021	Includes blower, wiper and soft cloth.	Difficult to remove sticky substances.
4	Electrostatic dust removal using charge induction.	Sreedath panat, kripa k varanasi	2022	A charge applied on the panel repels the dust	Not applicable for all type of dusts.
5	Design and Development of an Autonomous Solar Panel Cleaning System	Nurhasliza Hashim	2020	Experimental design and testing of a robotic cleaning system	Cleaning schedules and water conservation.

Literature Review

Sl. No.	Paper title	Author name	Year	Key Points	Gaps Identified
6	A Control Strategy for an Autonomous Robotic Vacuum Cleaner for Solar Panels	G. Aravind; Gautham Vasan; T.S.B. Gowtham Kumar; R. Naresh Balaji; G. Saravana Ilango	2014	• Lightweight design, ensuring optimal operation without causing damage.	• The vacuum cleaner need frequent maintanance
7	A Comparative Study of Dust Cleaning Methods for the Solar PV Panels	Mallikarjun Hudedmani,Gita Joshi,R M Umayal,Ashwini Revankar	2017	Dust accumulation on solar PV panels significantly reduces energy efficiency, making regular cleaning essential to maintain optimal performance.	-
8	Design and Development of a Cleaning Robot for Solar Panels with Sun Tracking	Wallaaldin Eltayeb,Yedukondalu Gandham,Dr. A. Srinath	2021	Experimental testing and comparison of cleaning methods. Identification of effective cleaning methods for dusty environments.	Cleaning system uses water. Impact of Sun light

Survey at SJEC





Survey at NMPT

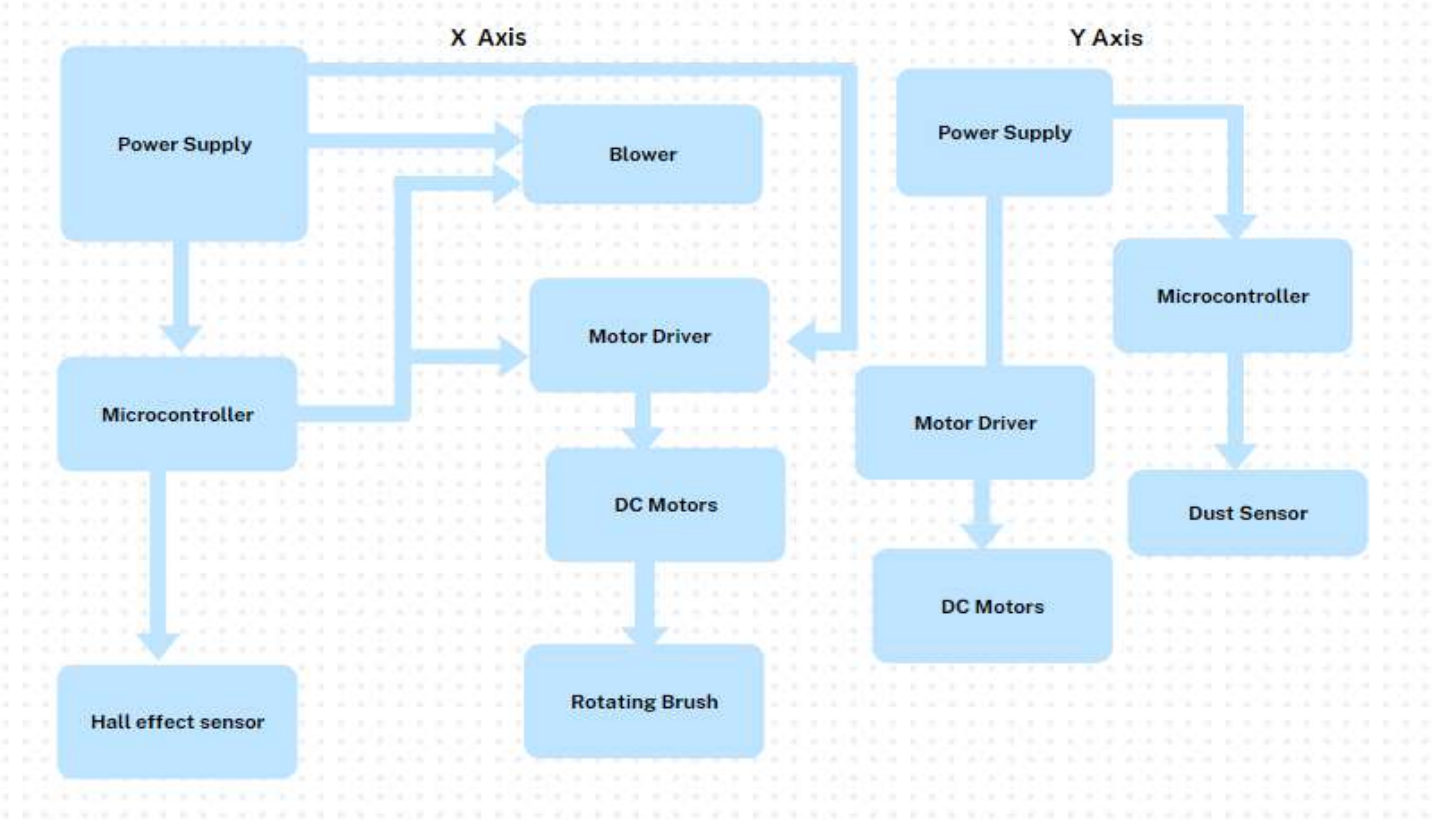




Objectives

- Make the cleaning of the solar panels simple and automated thereby reducing human intervention and labour cost.
- Develop an IoT enabled prototype to control the system and get the real time data using a mobile app.
- Integrate IoT and X and Y axis movement to ensure efficient functioning.

Methodology

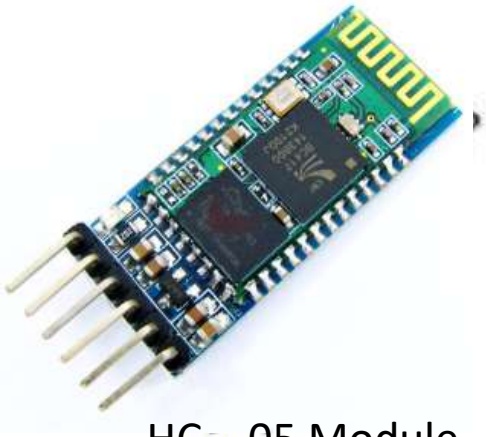


Block diagram

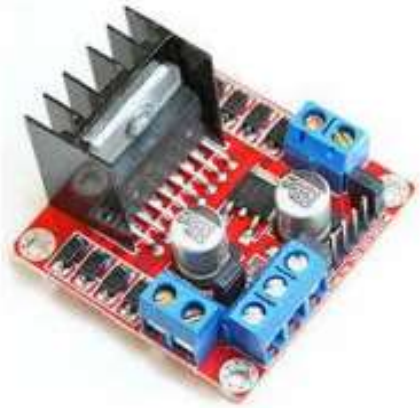
Methodology

- We conduct a qualitative analysis on available solar panel cleaning techniques and develop a water-free cleaning method that integrates a blower , brush.
- We confirm the technical feasibility of this waterless cleaning method and select components with the required specifications to build a prototype.
- We build a prototype using the selected components and evaluate its performance.
- Based on the test outcomes, we finalize the design of our water-free solar panel cleaning system

Components



HC – 05 Module



L298N Motor Driver



MT3608 module



Hall Effect Sensor Module A3144EUA

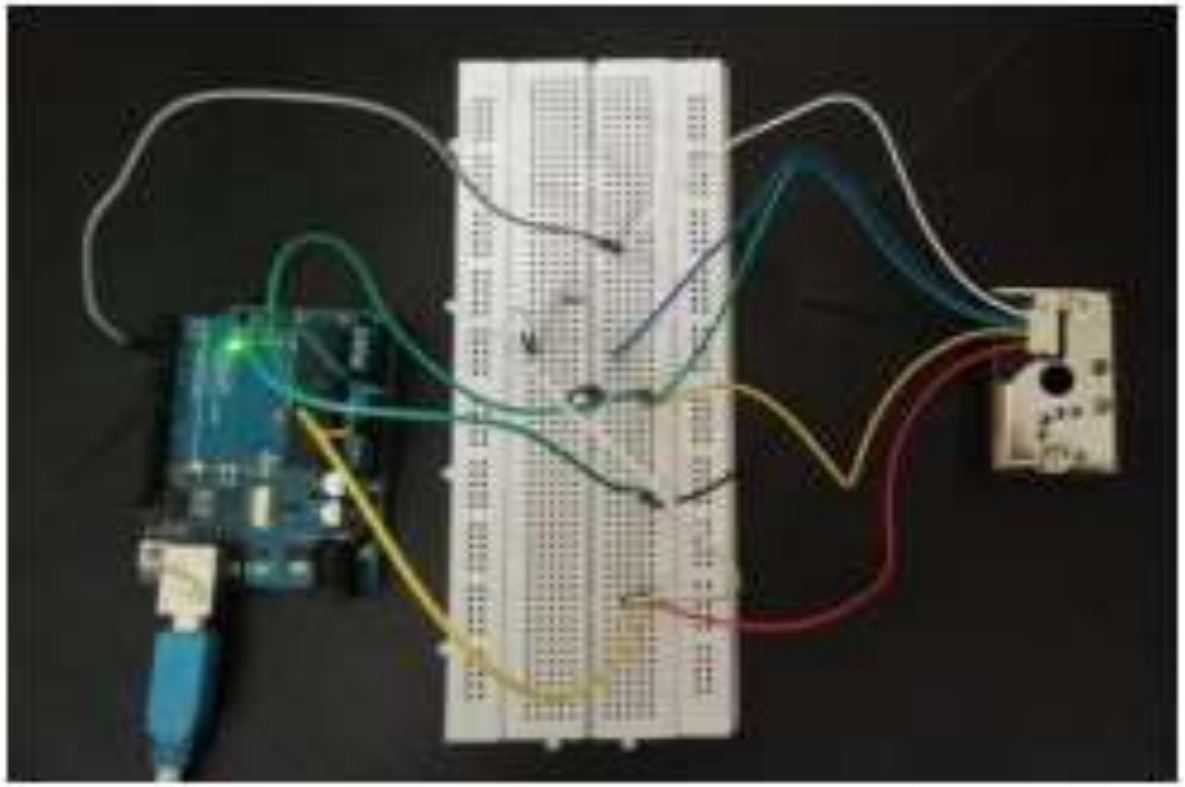


12V DC Blower Fan



DC Motor

Components



Dust sensor Integration with Arduino UNO

Results and Discussion

- The project is 85% complete, with a functional prototype developed.
- Developed a user-friendly app interface for our project using MIT App Inventor.
- This application serves as the central control system, allowing users to manage various ON/OFF functions with ease and also get real time dust intensity data.
- The prototype has successfully achieved two axis movement on the solar panel surface.
- However, some improvements are yet to be done on the cleaning mechanism.

Results and Discussion

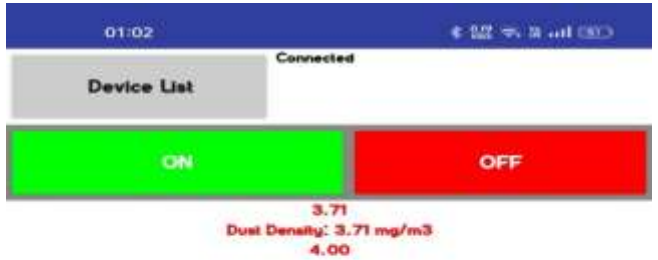


Design Implementation

Results and Discussion



App user interface



App control panel interface

Conclusion

- The model successfully accomplishes the one axis movement of the station upon the solar panel that performs the cleaning process.
- The use of HC 05 Bluetooth module enables the user to control the model automatically to some extent.
- Aluminum is used as the frame because of its light weight and low cost.

Future scope And Implications

- More efficient material can be developed that can replace aluminum as the frame for the model.
- Weather sensors can be used to improve the efficiency by enable the working of the model only when necessary
- A system can be developed that can provide power supply to the model harnessing the solar energy from the same panel.
- Mode of operation can be subdivided into many levels depending on the dust concentration on the panel.

Participation in competitions



ST JOSEPH ENGINEERING COLLEGE
AN AUTONOMOUS INSTITUTION
Vamanjoor, Mangaluru - 575 028

Department of Electronics and Communication Engineering

In association with

ISTE SJEC Student Chapter

Mini Project Exhibition 2024

This is to certify that the Best Project Prize in the Mini Project Exhibition held on 01 August 2024 at SJEC was awarded to the project work titled Dry Solar Panel Cleaner

carried out by the following students of BE in ECE

- Akarshan V H
- B Vikas
- Dennith Hegde
- Manish

under the guidance of Ms Vijayalaxmi H. M

Ms Simi P Thomas & Ms Vinitha Pasanha
Coordinators - Mini Project Exhibition 2024

Dr Dayakshini
HOD - ECE Department

Link to the project

Participation in competitions



innovation Cell www.snmpolytechnic.org

Navathvam 2024

Date:

To,
The Organizing Committee,
Navathvam-2024,
S N Mudbidri Polytechnic, Moodubidire

Subject: Authorization letter for participation in Navathvam-2024

Dear Sir/Madam,

The present communication functions as a formal approval for the participation of the students and faculty members mentioned below from **St Joseph Engineering College** in the next **Navathvam-2024** event, scheduled to occur on 13.09.2024 at **S N Mudbidri Polytechnic.**

The following are the participating students and faculty members:

Project Title	: Dry Solar Panel Cleaning System
Team leader(Member 1)	: B Vikas
Email Id	: vikasbantaje@gmail.com
Member 2	: Deevith Hegde
Member 3	: Akarshan V H
Member 4	: Manish
Accompanying Faculty	: Ms Vijayalakshmi H Manjunatha



innovation Cell

INSTITUTION'S INNOVATION COUNCIL

1984 MOODBIDRI

CERTIFICATE

OF APPRECIATION

This is to certify that,

Deevith Hegde

*from **St Joseph Engineering College** has actively participated in the **Technical Project Exhibition "Navathvam-2024"** organized by IIC of SNMP on 13th September 2024 at S N Mudbidri Polytechnic, and presented the project titled **"Dry Solar Panel Cleaning System"** in the category of **"e-Sparx"**.*

NAVATHVAM 2024

Convener
IIC of SNMP

Principal
President, IIC of SNMP

Ref. No.: SNMP/N-2024/85-17-02

References

i. Innico.R, Karthish Durai.A.M, Poovalingam. M ,“Automatic solar panel cleaning using wiper”, IJMR(International journal of multi disciplinary research) 2023.

ii. Santosh Kumar S, Shivashankar, Keshava Murthy “Solar Powered PV Panel Cleaning Robot”, International Conference on Recent Trends on Electronics, Information, Communication & Technology,2020.

iii. Md Rawshan Habib, Shahnewaz Tanvir “Automatic Solar Panel Cleaning System Based on Arduino for Dust Removal” International Conference on Artificial Intelligence & Smart Systems, 2021

iv. Sreedath Panat , Kripa K Varanasi , “Electrostatic dust removal using adsorbed moisture assisted charge induction”, IEEE Journal of Photovoltaics,2022.

v. Paper published by The World Economic Forum 2022.

THANK YOU